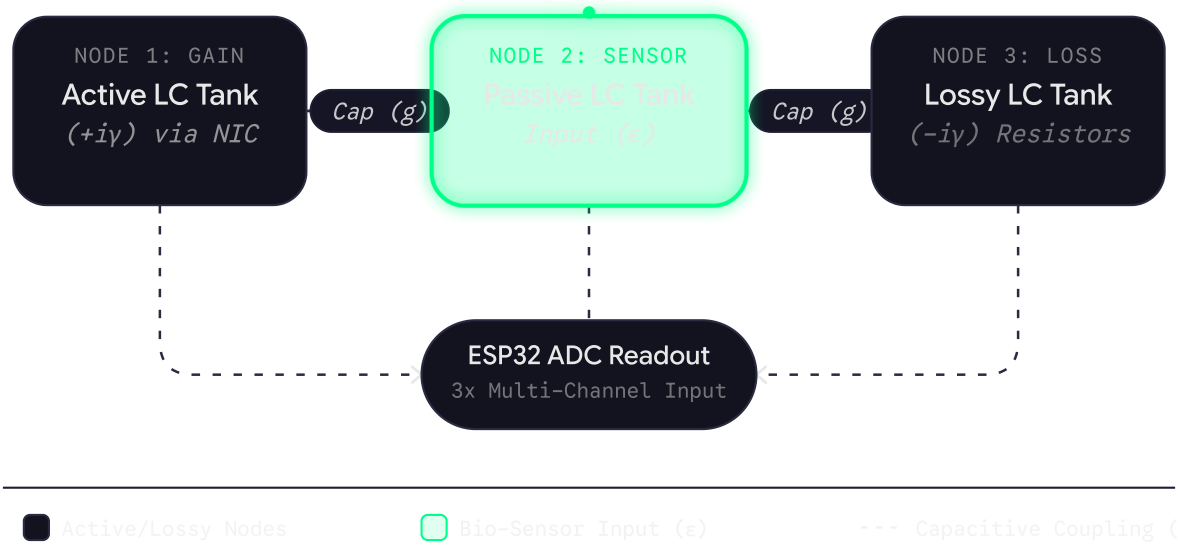
ATLAS • EP3 LAB • #020208 LIVE • 3-NODE

3-Resonator EP3 Circuit Architecture

Exceptional Point 3 (EP3) non-Hermitian sensor topology. Hover/select nodes to view matrix term. Bio-Sensor Input node exhibits cubic-root sensitivity $\Delta\omega \propto \varepsilon^{1/3}$.



Node 2: Bio-Sensor Input (Passive)

$H[2,2] = \omega_o + \varepsilon$

$$H_{EP3} = \begin{bmatrix} \omega_o + i\gamma & g & 0 \\ g & \omega_o + \varepsilon & g \\ 0 & g & \omega_o - i\gamma \end{bmatrix}$$

Central perturbation site (ε). Perturbation in this node shifts eigenvalues with ultra-high cubic-root sensitivity $\Delta\omega \propto \varepsilon^{1/3}$ at EP3.

CUBIC-ROOT ENHANCEMENT ACTIVE • $\Delta\omega \propto \varepsilon^{1/3}$ • 10× SNR @ EP3

H[1,1]=Ω₀+iΓ • H[2,2]=Ω₀+E • H[3,3]=Ω₀-iΓ • G = CAP EP3 • ATLAS

EP3 SENSOR — INTERACTIVE VISUAL — BIO-SENSOR INPUT #020208 #e8e8e8 #00ff88